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Collagen is one of the most promising materials used for scaffolds for tissue repair. In addition to its most advantageous mechanical and structural characteristics, it is notable that the collagen scaffold promotes cell attachment and migration and allows preservation of a phenotype of specialized cells. Moreover, such a scaffold is an important device for the delivery of growth factors that promote tissue repair into a site of injury. The long-term objective of this project is to engineer a collagen-like molecule with novel structural and biological characteristics and use this protein to produce materials for cartilage repair. The overall hypothesis is that attachment and migration of chondrocytes through three-dimensional matrices depends on a mechanism that involves an interaction of integrins with a specific region in collagen II, and that this interaction is critical for synthesis of correct tissue. In addition, we hypothesize that the high-affinity interaction between bone morphogenetic protein 2 (BMP-2) and collagen II is limited to a specific region of the collagen monomer. In these studies, we propose to use genetically engineered collagen II variants with deleted

Task Description:	D-periods to analyze critical sites in collagen II in detail. Characterization of these sites will provide a basis for the rational design of "super collagen" scaffolds, in which critical sequences of collagen II will be multiplied to provide a higher density of the specific sites of interaction. Such a high density of the most important collagen fragments will promote a rapid and uniform population of a scaffold with seeded cells and will contribute to faster repair of cartilage. The specific aims are: (1) To determine how the specific regions of collagen II regulate the attachment and spatial distribution of chondrocytes seeded onto a scaffold. (2) To identify cell receptors that interact with the D4 period of collagen II. (3) To design a genetically engineered collagen for adhesion of cells and growth factors. Because of their mechanical and biological characteristics, collagens have numerous biomedical applications. In tissue engineering, for example, collagens are employed as material for matrices that promote tissue repair by supporting cell growth and by facilitating delivery of growth factors to a site of injury. Almost all of the collagen available on the market today is obtained from bovine hide and bones. These sources allow for the extraction of large quantities of collagenous proteins in a cost-effective manner, but, at the same time, they do not permit molecular-level modifications that could enhance biological or mechanical characteristics of the product. Expressing recombinant collagens circumvents this problem and offers a significant potential for the rational engineering of collagenous proteins. Such rational engineering of collagens with enhanced biological properties would generate products for use in tissue engineering, fibrosis, wound healing, and drug and cell delivery. Successes in expressing recombinant fibrillar collagens in mammalian cells prompted the development of market-oriented, large-scale production technologies and, at present, recombinant collagens are expressed in bacteria, yeast, insect cells, transgenic animals, and transgenic plants. All these technologies, however, focus on expressing the recombinant collagens identical to those extracted from animal tissues, and to date, rational engineering of collagens with specifically assigned biological characteristics has not been addressed. Because most biological processes in which fibrillar collagens take part depend on site-specific interactions, the concept of engineering collagen-like proteins that consist of repeats of a collagen domain critical for particular site-specific function is very important. Such rational engineering would create a custom-tailored “super-collagen” with enhanced specific biological properties and characterized by a high density of the active sites. Moreover, by adding or removing specific domains it will be possible to modulate collagen-dependent signaling and to influence stability and self-assembly of collagen molecules. Such collagen-like proteins have the potential to be used in “smart-scaffolds” that encode specific commands for controlling tissue formation. It is not known, however, whether engineered collagen ? chains consisting predominantly of multiplied specific domain but lacking most of other domains found in native collagen would fold into a thermostable collagen triple helix, a structure critical for biological and mechanical functions. The task objective is to engineer collagen-like proteins for novel scaffolding materials containing a high density of sites critical for interaction with cells. The results of studies will provide a basis for the rational engineering of novel collagen-like proteins customized for specific biomedical applications. Cartilage is an important target for tissue engineering. It has been established that the following elements are required for successful cartilage engineering: (i) cells, (ii) growth factors, and (iii) scaffold. Because the proposed research focuses on a detailed analysis of the role of collagen II in interaction with chondrocytes and with growth factors used to promote cartilage regeneration, it addresses all critical elements of cartilage engineering. This research will yield an understanding of the mechanism of attachment and migration of chondrocytes through collagenous matrices designed to promote cartilage repair. Moreover, this research will provide important information about the interaction of collagen with growth factors belonging to the TGF-beta family. The results of the proposed research will be used for a rational design of genetically engineered "super collagen" molecules with improved abilities to promote cell growth and binding of growth factors. In addition, combining the "super collagen" with nanofibrillar or sponge-like matrices will allow creation of a uniform and controllable architecture of scaffolds for cartilage repair. The central characteristic of experimental design of the research proposed in this application is the use of the genetically engineered collagen II to map its sites critical for biologically relevant interactions. In the past, other approaches to understand the function of discrete collagen domains were attempted. In particular, fragments obtained by the chemical cleavage of collagen with cyanogens, linear peptides with amino acid sequences derived from collagens, were used in binding studies. In these experiments, however, the collagen fragments were denatured and, as a result, lacked the native triple-helical conformation. In the experiments with collagen I, attempts were made to localize the integrin binding sites by using synthetic, collagen-like peptides that were triple helical because they were stabilized by an addition of Gly-Pro-Hyp repeats. Some of these peptides were active in promoting fibroblast adhesion. However, they were not able to compete for the binding of the native collagen I. These and other observations indicate that the biologically active sites in collagens are best studied by assaying their role in various interactions to intact monomers or partially deleted derivatives of the intact monomers in which the binding sites are present in the normal context of amino acid sequences and three-dimensional conformation. The cassette system we developed makes detailed studies of the role of collagen II domains in interactions with cells and macromolecules possible. This system will be critical not only for generation of the artificial collagen matrices for tissue engineering but, in a broad sense, will also serve as a tool for the development of “custom collagens” for a number of medical applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research on rationally engineered collagen-like proteins aims into development novel materials for use in various biomedical applications. Development of novel, custom tailored collagenous proteins will have great positive impact on tissue and organ engineering, drug delivery, tissue repair, and inhibition of fibrosis.
Task Progress:	During the last year, work has progressed in the following areas: (i) engineering and characterization of the multi-D1, D2, D3, and D4 collagen variants, (ii) growing chondrocytes in scaffolds coated with the multi-D4 collagen, (iii) mapping sites in collagen II critical for interaction with the discoidin domain receptors (DDR). (i) Engineering of the Multi-D1, D2, D3, and D4 Collagen Variants -- Employing a procollagen II DNA cassette system, we assembled DNA constructs encoding recombinant collagen II variants that consist of multiplied D1, D2, D3, or D4 periods of collagen II. The DNA constructs were expressed in HT-1080 cells. The novel collagen-like proteins were characterized for their ability to form thermostable triple helices. We demonstrated that the recombinant novel collagen-like proteins form stable triple-helical structures, thus, they can be used in artificial matrices. (ii) Growing Chondrocytes in Scaffolds Coated with the multi-D4 Collagen –We prepared scaffolds coated with genetically engineered multi-D4 collagen. The scaffolds were used to support growth of chondrocytes. After six weeks of culture, the constructs were analyzed with use of histology techniques. We demonstrated that the scaffolds coated with the multi-D4 collagen provide an excellent support for seeded cells. The number of cells that populated the scaffolds’ cavities was significantly higher than the number of cells seeded on uncoated scaffolds or the scaffolds coated with wild type collagen II. These results show that in comparison to tissue-derived collagen II the rationally engineered

	collagen-like protein has superior properties making this protein an excellent material for cartilage repair applications. At present, we continue to evaluate the multi-D4 collagen-coated scaffolds in long-term cell culture experiments. (iii) Mapping sites in collagen II critical for interaction with the discoidin domain receptors (DDR) –Discoidin domain receptors are, uniquely among tyrosine kinase receptors, activated by collagen. We demonstrated that the DDR2 is specifically activated by collagen II. We also identified the region of collagen II that is critical for activation of the DDR2. These results indicate that activation of DDR2 is site-specific.
Bibliography Type:	Description: (Last Updated: 06/23/2004)
Articles in Peer-reviewed Journals	Majsterek, I., McAdams, E., Adachi, E., Dhume, S.T., Fertala, A. "Prospects and limitations of the rational engineering of fibrillar collagens." Protein Science 12:2063-2072 , Jan-2003
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